

METIMMOX-1 Clinical Effectiveness Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This report presents clinical effectiveness results for the METIMMOX-1 trial, comparing biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer.

2 Results

2.1 Cox Proportional Hazards Models

Cox proportional hazards regression models were fitted comparing each biomarker-defined treatment subgroup to a common reference of all patients receiving standard of care (Rx=0), adjusting for age and sex. This approach pools all control patients regardless of biomarker status as the reference group.

2.2 Median Survival Times and Overall Outcomes

2.3 Clinical Effectiveness Summary

2.3.1 Table 1: Overall Survival and Quality-Adjusted Outcomes by Strategy and Biomarker Subgroup

Note: All outcomes are undiscounted (discount rate = 0%). All results are derived from parametric survival models. Overall strategy outcomes are weighted by biomarker prevalence. Subgroup QALYs are calculated from the health economic model accounting for time in progression-

Table 1: Clinical Effectiveness Results (Undiscounted)

Strategy and Subgroup	Median PFS (months)	Median OS (months)	Mean Life Years	Mean QALYs
Standard of Care				
Control (FLOX alone)	11.74	22.78	2.09	1.80
CRP-guided Strategy				
CRP-guided (overall)	13.12	24.17	2.23	1.92
CRP-positive (Rx=1)	16.80	27.16	2.50	2.13
CRP-negative (Rx=0)	11.74	22.78	2.09	1.80
TLR-guided Strategy				
TLR-guided (overall)	7.13	15.42	1.48	1.30
TLR-positive (Rx=1)	9.44	19.33	1.78	1.55
TLR-negative (Rx=0)	4.37	10.36	0.94	0.84
TMB/BRAF-guided Strategy				
TMB/BRAF-guided (overall)	13.12	23.01	2.11	1.82
TMB/BRAF-positive (Rx=1)	15.42	23.25	2.14	1.85
TMB/BRAF-negative (Rx=0)	11.74	22.78	2.09	1.80

free and progressed health states with different utility weights. Rx=1 indicates treatment with alternating FLOX/nivolumab; Rx=0 indicates FLOX alone (standard of care).

2.3.2 Table 2: Hazard Ratios for Progression-Free Survival by Biomarker Status

Table 2: Hazard Ratios for Progression-Free Survival by Biomarker Status

Strategy and Biomarker Subgroup	HR for PFS (95% CI)
Standard of Care	
Control (FLOX alone)	Reference
CRP-guided Strategy	
CRP-guided: Biomarker-positive (Rx=1)	0.45 (0.20-1.00)
CRP-guided: Biomarker-negative (Rx=0)	1.95 (1.00-3.82)
TLR-guided Strategy	
TLR-guided: Biomarker-positive (Rx=1)	0.46 (0.22-0.97)
TLR-guided: Biomarker-negative (Rx=0)	3.66 (1.68-7.98)
TMB/BRAF-guided Strategy	
TMB/BRAF-guided: Biomarker-positive (Rx=1)	0.45 (0.19-1.04)
TMB/BRAF-guided: Biomarker-negative (Rx=0)	1.62 (0.83-3.14)

Note: Hazard ratios compare each biomarker-defined treatment subgroup to all patients receiving standard of care (Rx=0), pooled regardless of biomarker status, adjusted for age and sex. $HR < 1$ indicates better survival than pooled control group.

2.3.3 Table 3: Hazard Ratios for Overall Survival by Biomarker Status

Table 3: Hazard Ratios for Overall Survival by Biomarker Status

Strategy and Biomarker Subgroup	HR for OS (95% CI)
Standard of Care	
Control (FLOX alone)	Reference
CRP-guided Strategy	
CRP-guided: Biomarker-positive (Rx=1)	0.73 (0.37-1.46)
CRP-guided: Biomarker-negative (Rx=0)	1.97 (1.05-3.69)
TLR-guided Strategy	
TLR-guided: Biomarker-positive (Rx=1)	0.79 (0.41-1.53)
TLR-guided: Biomarker-negative (Rx=0)	2.02 (1.05-3.89)
TMB/BRAF-guided Strategy	
TMB/BRAF-guided: Biomarker-positive (Rx=1)	0.86 (0.42-1.76)
TMB/BRAF-guided: Biomarker-negative (Rx=0)	1.45 (0.79-2.64)

Note: Hazard ratios compare each biomarker-defined treatment subgroup to all patients receiving standard of care (Rx=0), pooled regardless of biomarker status, adjusted for age and sex. HR < 1 indicates better survival than pooled control group.

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